RouteRec: Behavior-Guided Expert Routing for Sequential Recommendation

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Abstract

Sequential recommendation is commonly optimized with one shared computation pipeline, even though interaction regimes vary substantially across users, sessions, and time scales. In practice, some prefixes are dominated by stable preference, some by repetition, and others by local exploration. This mismatch between behavioral diversity and shared computation is increasingly visible as backbones become stronger: performance improves, but the model still offers little control over *how* different behavior patterns should trigger different computation.

Mixture-of-experts (MoE) is a natural candidate for conditional computation, yet many sequential MoE variants still route primarily from hidden states. This creates a semantic bottleneck: the same latent signal is asked to encode sequence content and decide routing allocation. As a result, expert assignment is often effective but opaque, making it difficult to interpret specialization, diagnose failure modes, or align routing with observable behavioral factors.

We propose **RouteRec**, a behavior-guided MoE framework that explicitly decouples routing signals from representation signals. RouteRec keeps a strong sequence encoder for hidden-state construction, while routing is driven by compact behavior cues extracted from standard logs (timestamps, optional categories, and interaction statistics). These cues are organized into four semantic families (Tempo, Focus, Memory, Exposure) and consumed through three temporal scopes (macro, mid, micro), so the routing structure matches evidence granularity rather than imposing a single gating bias everywhere.

This design is intentionally lightweight: we avoid heavy side-information modeling in the routing branch and instead emphasize interpretable, transferable summaries that are easy to compute across datasets. We further provide a dedicated preliminary formulation to make the method readable without overloading notation. Experiments show that RouteRec improves next-item recommendation over strong baselines, and diagnostic analyses indicate that improvements are associated with more behavior-consistent expert usage rather than unstructured parameter scaling.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

sequential recommendation, mixture of experts, conditional computation, routing, behavioral features

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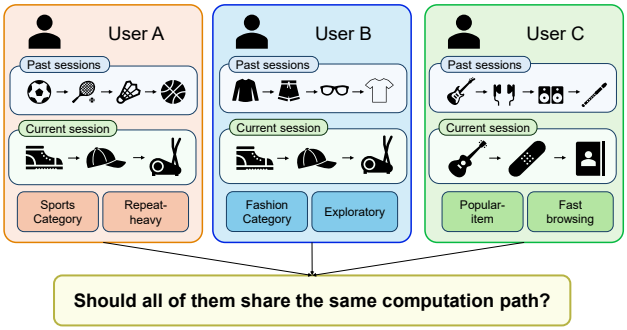

Figure 1: Different users and sessions can require different computational paths.

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1 Introduction

Sequential recommendation is usually formulated as next-item prediction over an ordered interaction history. At first glance, this appears to be a homogeneous prediction task, but real user behavior is not homogeneous at all. Even for the same user, one session may reflect persistent preference, another may reflect short-term mission behavior, and another may be dominated by quick exploratory clicks. Prior surveys and studies consistently describe this multi-regime nature of sequential and session-based recommendation [29, 30].

Modern backbones have become very strong at sequence modeling [8, 11, 14, 17, 21, 24], but they still predominantly rely on shared transformations across behaviorally different cases. This design is efficient and often performant on average, yet it leaves an important modeling question open: when two prefixes imply fundamentally different behavioral contexts, should they really invoke the same computation path?

Figure 1 provides an intuitive view of this gap. Distinct users and sessions can exhibit markedly different tendencies in pace, repetition, and local exploration, but a one-path design does not expose a mechanism for behavior-specific computation allocation. In other words, representation quality improves, but routing semantics remain implicit.

Mixture-of-experts (MoE) provides a direct mechanism for conditional computation [5, 9, 10, 23]. However, in many recommendation settings, routing is still derived from hidden states. This is practical, but it also entangles two roles that are

better separated: representing sequence content and deciding where computation should go. When both are handled by the same latent signal, expert specialization becomes harder to interpret and debug.

This limitation becomes more serious when the goal is not only accuracy but also controllability and diagnosis. If routing is purely latent, it is difficult to answer basic questions such as: Which behavior pattern activated which expert? Are similar sessions routed consistently? Is an expert specializing in a meaningful regime or merely in an embedding artifact? These questions are central for robust deployment and scientific understanding, but are weakly supported by hidden-state-only routing.

We therefore adopt a simple principle: **decouple routing signals from representation signals**. In RouteRec, hidden states remain the content channel processed by experts, while routing is controlled by compact behavior cues traceable to logs. This separation preserves backbone strength while making routing decisions easier to inspect and reason about.

RouteRec also aligns routing with temporal scope. Behavior evidence near the prediction point is not equivalent to long-range history evidence; forcing a single routing bias for all horizons is unnecessarily rigid. We therefore use a multi-stage design with macro, mid, and micro routing, allowing coarse-to-fine expert allocation that reflects cross-session tendency, within-session evolution, and immediate transitions.

Our goal is not to maximize notation complexity or introduce heavy side branches. Instead, we emphasize a compact, transferable routing view built from broadly available signals. This makes the approach practical across datasets and improves method readability, while still enabling richer conditional computation than shared-path baselines.

This paper also introduces a dedicated *Preliminary* section before the main method. The goal is pragmatic: define the problem statement, notation, and MoE basics once, so the method section can focus on RouteRec-specific design choices.

The main contributions of this work are as follows:

- We identify a core limitation of hidden-state-only routing in sequential MoE recommendation and formalize routing-representation decoupling as a practical design principle.
- We introduce RouteRec, a behavior-guided multi-stage MoE framework that routes with compact cues from standard interaction logs instead of hidden-state-only gating.
- We provide a clear preliminary formulation (problem statement and MoE background) to support readable method development without excessive symbol overhead.
- We show that RouteRec improves recommendation quality over strong baselines and yields more interpretable, behavior-consistent routing patterns in analysis.

2 Related Work

2.1 Sequential Recommendation Backbones

Sequential recommendation has progressed from recurrent and convolutional models to graph and Transformer architectures [8, 11, 25, 27, 31]. Recent methods further strengthen performance with time-aware attention, contrastive objectives, and structural inductive bias [4, 14, 17, 21, 24, 28, 32]. These advances improve representation quality, but they generally retain shared computation across behaviorally different cases.

2.2 Heterogeneous User State Modeling

Many studies explicitly acknowledge non-stationary behavior, including short-term intent modeling and multi-interest extraction [3, 12, 13, 16, 18, 35]. Most approaches handle heterogeneity inside representation learning, for example by attention reweighting or interest decomposition. RouteRec is complementary: it treats heterogeneity as a routing problem and uses explicit behavior cues to modulate computation paths.

2.3 Feature-Aware Sequential Recommendation

Feature-aware methods use timestamps, side information, and feature channels to strengthen sequence encoding [4, 14, 33, 37]. In most cases, these signals are fused into hidden representations. RouteRec uses a different role assignment: features are primarily routing controls, while representation transformation remains expert-specific over hidden states.

2.4 MoE in Recommendation

Classical and modern MoE work demonstrates the value of conditional computation [5, 9, 10, 23]. Recommendation-oriented expert architectures also show strong capacity benefits [1, 15, 19, 26, 36]. Our distinction is not simply adding experts, but making routing semantics behavior-grounded and interpretable through explicit cue design.

3 Preliminary

3.1 Problem Statement

For each user, we observe interactions ordered by time and train the model for next-item prediction. In one training instance, we denote the observed prefix as $x_{1:t}$ and the target as x_{t+1}. Each interaction includes at least an item identity and timestamp, and may additionally include coarse category information depending on the dataset.

To keep the formulation readable, we use two signal channels throughout this paper. The first is the representation channel, where a sequential encoder maps observed prefixes to hidden states $\mathbf{h}_t$. The second is the behavior channel, where lightweight statistics from the same prefix are summarized into a compact cue vector $\mathbf{b}_t$. RouteRec uses $\mathbf{h}_t$ for expert transformation and $\mathbf{b}_t$ for routing decisions. This separation

is the central modeling choice that the later method section instantiates stage by stage.

The behavior channel is intentionally simple: cues are derived from standard log attributes rather than from a deep side encoder. This keeps routing signals stable across datasets and easier to interpret during analysis. In practice, cues capture aspects such as pace, repetition tendency, focus concentration, and popularity exposure. We do not force a large handcrafted taxonomy in this section; instead, we keep the abstraction compact and defer stage-specific feature construction details to Method.

3.2 MoE Background for Sequential Models

In a standard MoE layer with E experts, an expert-weighted output at position t is

$$\tilde{\mathbf{h}}_t = \sum_{e=1}^{E} \pi_{t,e} \operatorname{Expert}_e(\mathbf{h}_t), \tag{1}$$

where $\boldsymbol{\pi}_t \in \mathbb{R}^E$ is a routing distribution with $\sum_e \pi_{t,e} = 1$.

In many practical systems, $\boldsymbol{\pi}_t$ is computed from hidden states. This is effective but often semantically opaque, because routing and content encoding are entangled in the same latent space. RouteRec follows a different view: routing is computed from behavior cues, while hidden states are dedicated to expert transformation. The distinction is conceptually small but operationally important for interpretability and diagnosis.

Another practical point is temporal scope. Sequential recommendation combines long-range tendency and short-range fluctuation, so a single routing granularity can be suboptimal. RouteRec addresses this by applying the same MoE interface at multiple stages with different evidence windows. This keeps notation light while allowing routing behavior to adapt across horizons.

In the final model used in our main experiments, this multi-stage principle is realized with a shared router composition across all stages, while stage differences are introduced through temporal evidence windows and routing granularity. The next section formalizes this design.

4 Method

4.1 RouteRec Overview

Figure 2 summarizes RouteRec. We retain a strong sequential backbone and replace shared feed-forward computation with stage-wise MoE routing. The core interface follows the preliminary setup: hidden states are transformed by experts, and routing is controlled by behavior cues.

Let $\mathcal{T} = \{\text{macro}, \text{mid}, \text{micro}\}$ be the stage set. At stage $s \in \mathcal{T}$, routing uses a stage-specific behavior feature $\mathbf{f}_t^{(s)}$, while experts consume the shared hidden state $\mathbf{h}_t$:

$$\boldsymbol{\pi}_t^{(s)} = \operatorname{softmax}\Big(\operatorname{Router}_s(\mathbf{f}_t^{(s)})/\tau_s\Big), \tag{2}$$

$$\tilde{\mathbf{h}}_t^{(s)} = \sum_{e=1}^{E} \pi_{t,e}^{(s)} \operatorname{Expert}_e^{(s)}(\mathbf{h}_t). \tag{3}$$

Eq. (3) is the common stage interface. In the final RouteRec variant, all stages instantiate this interface with the same routing composition, while temporal differences arise from stage windows and routing granularity.

4.2 Behavioral Features from Raw Signals

RouteRec is designed for realistic data availability. It uses signals that are commonly present in interaction logs: item identity, timestamp, optional category, and optional corpus-level popularity. We intentionally avoid heavy side-information encoders in the routing branch. Instead, raw signals are converted into compact scalar summaries that are easy to compute, stable across datasets, and interpretable at analysis time.

For session s_m and target position t, the stage windows are

$$\mathcal{R}_t^{\text{macro}} = [s_{\max(1,m-H)}, \ldots, s_{m-1}], \tag{4}$$

$$\mathcal{R}_t^{\text{mid}} = [x_1, \ldots, x_{t-1}], \tag{5}$$

$$\mathcal{R}_t^{\text{micro}} = [x_{\max(1,t-w)}, \ldots, x_{t-1}], \tag{6}$$

where $H = 5$ and $w = 5$ in our main setting. A shared summary operator $\mathcal{A}(\cdot)$ maps each window to a low-dimensional feature vector:

$$\mathbf{f}_t^{(s)} = \mathcal{A}(\mathcal{R}_t^{(s)}), \qquad s \in \mathcal{T}.$$

The summary dimensions are grouped into four semantic families: *Tempo*, *Focus*, *Memory*, and *Exposure*. Tempo captures pace and interval statistics. Focus captures concentration and switching behavior over coarse categories or item groups. Memory captures recurrence, repetition, and overlap with recent context. Exposure captures interactions with popularity strata and their local drift. This decomposition is semantic rather than architectural: it gives routing modules an interpretable behavioral basis.

Importantly, the feature template is shared across stages, but each stage reads it through a different temporal lens. Macro features reflect persistent tendency and broad preference shape. Mid features track evolving intent inside the current session. Micro features emphasize immediate cues around the prediction target. This shared-template, stage-specific-view design keeps the feature branch compact while preserving temporal specificity.

4.3 Unified Router Composition

The final RouteRec architecture uses one shared scalar-group conditional composition at every stage. Experts are partitioned into G groups with C experts per group, so $E = GC$.

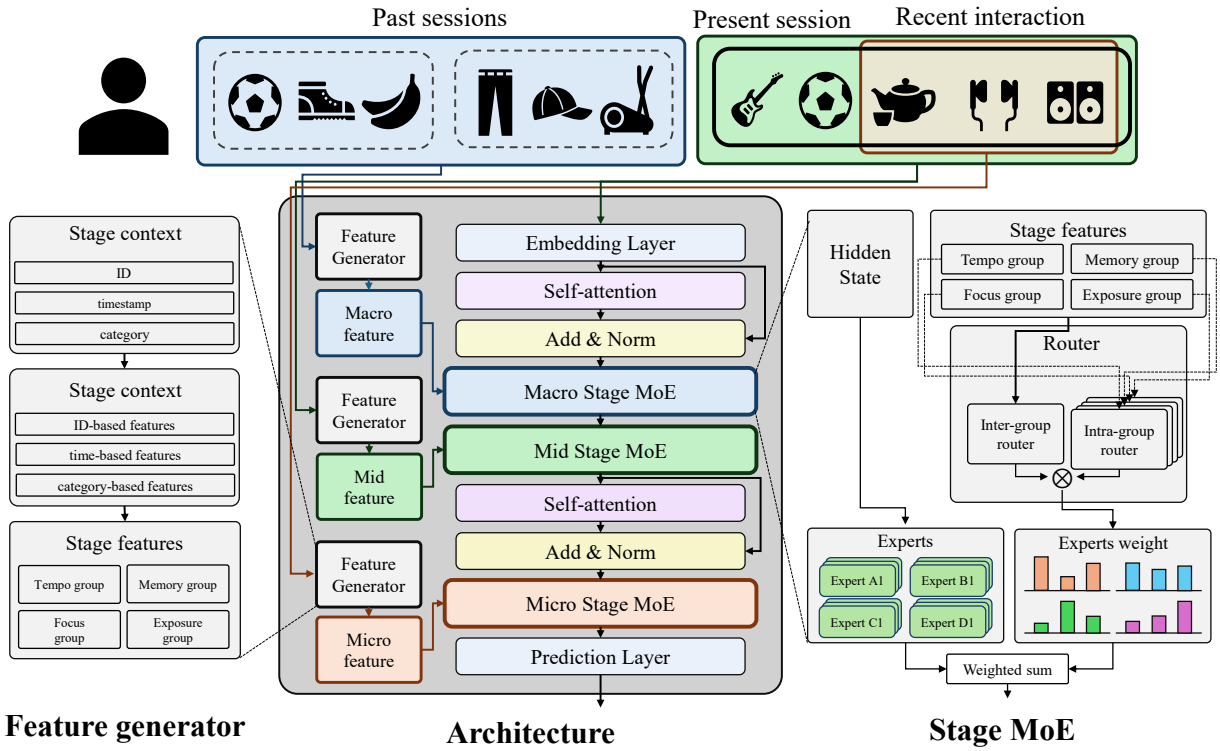


Figure 2: Overall RouteRec framework. (a) RouteRec keeps a strong sequential backbone and replaces shared feed-forward blocks with behavior-guided MoE stages. (b) All stages use the same scalar-group conditional routing composition: a group-level distribution and a conditional intra-group distribution are multiplied to form expert probabilities. (c) Macro, mid, and micro differ by temporal evidence scope and routing granularity, not by router formula.

For stage s and position t, the router produces

$$p_{t,g}^{(s)} = \mathrm{softmax}_g\left(z_{t,g}^{(s)}\right), \tag{7}$$

$$r_{t,g,c}^{(s)} = \mathrm{softmax}_c\left(u_{t,g,c}^{(s)}\right), \tag{8}$$

where $p_{t,g}^{(s)}$ is a group-level routing distribution and $r_{t,g,c}^{(s)}$ is a conditional within-group routing distribution. The final expert probability is their product:

$$\pi_{t,g,c}^{(s)} = p_{t,g}^{(s)}\, r_{t,g,c}^{(s)}, \tag{9}$$

followed by flattening (g, c) to expert index e. Eq. (9) is the same for macro, mid, and micro.

This unified composition is important for two reasons. First, it gives all stages a shared structural prior, which reduces stage-specific router engineering and simplifies debugging. Second, it keeps diagnostics comparable across stages because differences in routing behavior can be attributed to evidence scope rather than formula changes.

In RouteRec, routing is driven by behavior cues, while hidden states remain dedicated to expert transformation in Eq. (3). This preserves the routing-representation decoupling principle introduced in Preliminary.

4.4 Stage Instantiation by Temporal Scope

Although the router formula is shared, stages are not redundant. They differ in temporal evidence and routing granularity.

Macro and mid (session-level routing). Macro and mid compute one routing distribution per sequence from their respective windows, then broadcast it over valid positions inside the sequence. Macro uses broader cross-session history; mid uses within-session prefix evidence. This setting emphasizes stable allocation for slower-changing context.

Micro (token-level routing). Micro computes routing per token near the prediction target using short-horizon context. This enables fast adaptation to immediate transitions without changing the router form itself.

Formally, if Γ_s denotes the granularity operator at stage s, then

$$\boldsymbol{\pi}_t^{(s)} = \Gamma_s\left(\mathcal{R}_t^{(s)}\right),$$
$$\Gamma_{\text{macro}}, \Gamma_{\text{mid}} : \text{session-level},$$
$$\Gamma_{\text{micro}} : \text{token-level}.$$

Hence, RouteRec's coarse-to-fine behavior comes from stage evidence scope and update frequency, while the underlying router composition remains constant.

4.5 Training Objective

RouteRec is trained with a standard next-item objective and two router-side regularizers: route consistency and z-loss. The first regularizer shapes routing semantics, and the second stabilizes routing numerics.

Let $\mathbf{z}$ be final logits and y be the target item. The main loss is

$$\mathcal{L}_{\mathrm{CE}} = -\log \frac{\exp z_y}{\sum_j \exp z_j}.$$

For route consistency, we first compute a sample-level routing distribution at each stage by averaging token-level routing weights:

$$\bar{\boldsymbol{\pi}}_i^{\,s} = \frac{1}{L_i} \sum_{t=1}^{L_i} \boldsymbol{\pi}_{i,t}^{\,s}$$

where L_i is the valid sequence length. For each stage s, we construct a pair set $\mathcal{P}_s$ from feature-similar samples in the mini-batch and minimize

$$\mathcal{L}_{\mathrm{cons}} = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{|\mathcal{P}_s|} \sum_{(i,j) \in \mathcal{P}_s} \mathrm{JSD}\big(\bar{\boldsymbol{\pi}}_i^{\,s}, \bar{\boldsymbol{\pi}}_j^{\,s}\big)\,,$$

which encourages neighboring behavioral states to produce compatible routing distributions. In other words, semantically similar inputs should not induce arbitrarily different routing behavior.

To stabilize gating numerics, we adopt z-loss on router logits [38]:

$$\mathcal{L}_z = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{N_s} \sum_{(i,t) \in \mathcal{V}_s} \left(\log \sum_{e=1}^{E} \exp q_{i,t,e}^{\,s} \right)^2,$$

where $\mathcal{V}_s$ is the set of valid positions and $N_s = |\mathcal{V}_s|$. This term discourages excessive logit scale and helps prevent unstable, overly sharp routing. The final objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{cons}}\, \mathcal{L}_{\mathrm{cons}} + \lambda_z\, \mathcal{L}_z,$$

where λ_{cons} and λ_z are trade-off coefficients. This objective jointly optimizes prediction quality, semantic consistency of routing, and numerical stability of expert selection.

5 Experiments

5.1 Experimental Setup

We evaluate RouteRec on six public datasets. These datasets cover e-commerce, location recommendation, short-video consumption, music listening, movie preference, and anonymous clickstream behavior. The six corpora are Beauty, Foursquare, KuaiRec, LastFM, ML-1M, and Retail Rocket. Appendix A summarizes their provenance and processed statistics.

Across all datasets, interactions are first ordered chronologically and converted into session sequences. We then iteratively remove sessions shorter than 5 and items with frequency below 3, and split sessions into train/validation/test partitions in chronological order. The evaluation pipeline also cleans the held-out tail by re-forming contiguous validation and test segments and removing non-target unseen rows while preserving unseen targets, so that the final test setting remains strict but realistic. Beauty is derived from Amazon Beauty review logs and removes interactions with rating below 3. For KuaiRec and LastFM, we use compact filtered subsets to keep the experimental budget practical while preserving the underlying behavioral structure. When category or time signals are incomplete, RouteRec falls back to the same portable Tempo, Focus, Memory, and Exposure feature families computed from the available interaction log.

The baselines follow three groups. The first group contains conventional sequential recommenders, namely GRU4Rec, SASRec, and TiSASRec [8, 11, 14]. The second group contains recent strong backbones, including DuoRec, FEARec, BSARec, and SIGMA [4, 17, 21, 24]. The third group contains feature-aware or expert-related comparators, including DIFSR, FDSA, and FAME [15, 33, 37]. All methods are trained under the same optimization family with matched early stopping and comparable capacity ranges. Table 1 reports the compact main-text summary with HR@10, NDCG@10, and MRR@20, while Appendix A restores the full 3×3 cutoff grid. We use MRR@20 as the default scalar summary when a single validation or analysis metric is needed.

5.2 Overall Performance: Does Behavior-Guided Routing Help?

Table 1 presents the main overall comparison in the text. The selected metrics balance retrieval quality, rank sensitivity, and early precision, so the reader can assess the central result without scanning the full cutoff grid. RouteRec achieves the strongest pattern on Beauty, Foursquare, KuaiRec, and Retail Rocket, and remains competitive on LastFM and ML-1M, where strong shared-backbone baselines are already highly optimized.

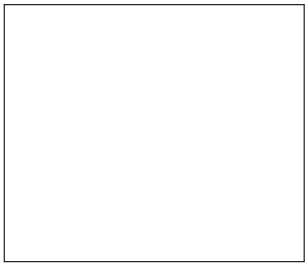
Two points are worth emphasizing. First, the clearest gains appear on Foursquare and Retail Rocket, where broader preference drift and sharp local intent shifts coexist. Second, the gain does not follow a simple capacity story: several baselines already use richer encoders or auxiliary objectives, yet the advantage emerges when computation is routed according to behavior rather than shared uniformly. The detailed @5/@10/@20 table is moved to Appendix A so that the main paper retains a readable experiments flow while still exposing the full result grid.

5.3 What Should Control Routing?

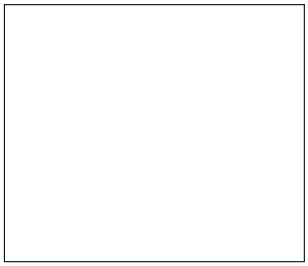
Figure 3 compares four routing controls: a shared FFN, hidden-only routing, mixed hidden+behavior routing, and the final behavior-guided router. Hidden-only routing can improve over a shared FFN, but the gain is smaller and less stable than behavior-guided routing. Re-injecting hidden states into the router also fails to produce a consistent advantage over the behavior-only design.

Dataset	Metric	GRU4Rec	SASRec	TiSASRec	DuoRec	FEARec	BSARec	DIF-SR	SIGMA	RouteRec	Improv.
Beauty	HR@10	0.0925	0.1082	0.1016	0.1187	0.1219	0.1168	0.0994	0.1048	**0.1256**	+3.0%
	NDCG@10	0.0592	0.0714	0.0667	0.0778	0.0798	0.0750	0.0661	0.0693	**0.0826**	+3.5%
	MRR@20	0.0548	0.0667	0.0619	0.0727	0.0741	0.0698	0.0623	0.0651	**0.0763**	+3.0%
Foursquare	HR@10	0.1827	0.2361	0.2210	0.2335	0.2178	0.2050	0.2106	0.2274	**0.2447**	+3.6%
	NDCG@10	0.1111	0.1527	0.1439	0.1494	0.1396	0.1301	0.1330	0.1464	**0.1589**	+4.1%
	MRR@20	0.0974	0.1378	0.1294	0.1336	0.1248	0.1168	0.1191	0.1315	**0.1430**	+3.8%
KuaiRec	HR@10	0.1732	0.1798	0.1647	0.1809	0.1796	0.1764	0.1691	0.1778	**0.1832**	+1.3%
	NDCG@10	0.1169	0.1236	0.1108	0.1241	0.1230	0.1208	0.1137	0.1216	**0.1264**	+1.9%
	MRR@20	0.1098	0.1160	0.1036	0.1164	0.1152	0.1133	0.1070	0.1141	**0.1185**	+1.8%
LastFM	HR@10	0.4016	**0.4456**	0.2564	0.2694	0.2774	0.4280	0.2585	0.4259	0.4305	-3.4%
	NDCG@10	0.3104	**0.3428**	0.2178	0.2236	0.2282	0.3345	0.2197	0.3383	0.3391	-1.1%
	MRR@20	0.2801	**0.3114**	0.2082	0.2126	0.2168	0.3045	0.2094	0.3071	0.3083	-1.0%
ML-1M	HR@10	0.1694	**0.1887**	0.1878	0.1652	0.1790	0.1841	0.1850	0.1790	0.1826	-3.2%
	NDCG@10	0.0908	0.0952	0.0953	0.0838	0.0869	0.0983	**0.0990**	0.0936	0.0978	-1.2%
	MRR@20	0.0711	**0.0783**	0.0781	0.0659	0.0679	0.0747	0.0756	0.0719	0.0741	-5.4%
Retail Rocket	HR@10	0.3779	0.4391	0.4205	0.4427	0.4418	0.4327	0.4400	0.4155	**0.4723**	+6.7%
	NDCG@10	0.2504	0.2907	0.2798	0.2921	0.2915	0.2870	0.2896	0.2748	**0.3094**	+5.9%
	MRR@20	0.2001	0.2348	0.2241	0.2362	0.2354	0.2278	0.2327	0.2163	**0.2495**	+5.6%

Table 1: Main overall results in the text. We report HR@10, NDCG@10, and MRR@20 in the main paper, and defer the full $k \in \{5, 10, 20\}$ grid to Appendix A. Bold and underlined values denote the best and second-best results.



(a) Avg. ranking quality across routing controls



(b) Route consistency over feature-similarity buckets

Figure 3: Routing-control comparison. (a) Average MRR@20 across shared FFN, hidden-only, mixed, and behavior-guided routing. (b) Routing similarity over feature-similarity buckets.

The consistency panel makes the structural reason clearer. When routing is tied to observable cues rather than hidden-state geometry alone, behaviorally similar cases are assigned more compatible expert patterns. The advantage of RouteRec therefore comes from grounding the routing decision in semantically readable evidence, not from adding a more complicated gate.

5.4 Why Is the Staged RouteRec Structure Effective?

Figure 4 examines the RouteRec architecture itself. Panel (a) removes macro, mid, or micro routing to isolate the role of each temporal scope. Panel (b) separates staged routing from generic FFN replacement by comparing dense replacement against one-, two-, and three-stage layouts. Panel (c) tests wrapper-map and stage-order variants to show that the final composition is not an arbitrary design choice.

Across all three views, the same pattern is expected. Performance rises when routing is distributed across temporal scopes, dense replacement alone does not recover the same gain, and the final wrapper map remains among the strongest choices. The benefit of RouteRec therefore comes from coarse-to-fine behavioral allocation rather than from expert insertion by itself.

5.5 Are Lightweight Cues Enough?

Figure 5 evaluates whether RouteRec depends on rich metadata or whether ordinary interaction logs already provide enough routing evidence. The left panel removes cue families from the full model, and the right panel normalizes each variant by the gain of the full-cue model. Removing category-derived or time-derived cues lowers performance, but the sequence-only variant still retains a substantial fraction of the improvement over the shared-backbone baseline.

This result strengthens the portability claim. RouteRec benefits from richer evidence when it is available, but it does not require a heavy side-information stack. A meaningful part of the routing gain can already be recovered from the signals that most sequential recommendation systems already log.

Detailed full-cutoff results and extended analyses, including slice-wise and transfer studies, are reported in Appendices A–F.

(a) Stage removal (b) Dense replacement vs. staged routing (c) Wrapper-map and order variants

Figure 4: Stage and layout ablations. (a) Stage removal. (b) Dense replacement versus staged routing. (c) Wrapper-map and stage-order variants.

(a) Avg. MRR@20 by cue family removal (b) Relative retention of the full-cue gain

Figure 5: Cue-family ablations. (a) Average MRR@20 under cue removal. (b) Gain retention relative to the full cue set.

6 Conclusion

RouteRec starts from a simple hypothesis: when behavioral evidence changes, the computation path should change as well. The experiments support this view from three main angles. First, RouteRec improves next-item ranking over strong sequential baselines across most datasets and cutoffs. Second, the gain comes from the routing signal and the staged design rather than from inserting experts indiscriminately. Third, lightweight cues derived from ordinary interaction logs already recover a substantial share of the benefit, which keeps the method practical even when side information is limited.

Taken together, these results suggest that compact behavioral cues provide a useful control interface for conditional computation in recommendation. Future work can build on this interface with broader transfer regimes, richer but still robust cue construction, and tighter deployment-aware constraints such as latency, continual adaptation, and router calibration.

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Dataset	Domain	Interactions	Sessions	Items	Avg. len.
Beauty	E-com.	33,488	4,243	3,625	7.9
Foursquare	POI	145,238	25,369	30,588	5.7
KuaiRec	Video	287,411	24,458	6,477	11.8
LastFM	Music	470,408	25,089	52,510	18.8
ML-1M	Movie	575,281	14,539	3,533	39.6
Retail Rocket	Browse	821,243	153,092	90,211	5.4

Table 2: Datasets used in the experiments.

A Dataset, Evaluation, and Full Results

This appendix summarizes the six datasets and the common evaluation protocol. The original releases are cited here for completeness [2, 6, 7, 20, 22, 34]. The dataset descriptions emphasize domain differences and signal availability because the routing branch depends on which behavioral evidence is exposed in each corpus.

A.1 Datasets and Protocol

Beauty and Retail Rocket stress sparse e-commerce revisit behavior, Foursquare and KuaiRec emphasize stronger short-horizon intent changes, and LastFM and ML-1M provide longer preference-driven histories. This spread is useful because it tests whether behavior-guided routing helps only in highly dynamic settings or remains competitive when preference structure is more stable.

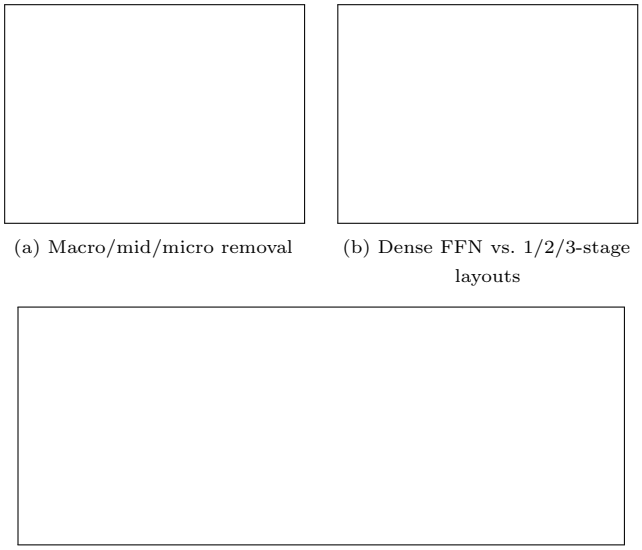
Signal availability also differs in a way that matters for routing. Beauty exposes item, time, and product-category cues after low-rating removal. Foursquare includes item, timestamp, and venue-category structure with bursty check-in dynamics. KuaiRec offers item, time, and coarse content-side metadata in short-video sessions with strong short-term intent shifts. LastFM provides repeated listening histories with artist-side signals, while ML-1M contributes longer preference-driven sequences with genre information. Retail Rocket is the sparsest side-information setting, relying mainly on anonymous event streams with item and time signals.

All datasets are sessionized from timestamp-ordered interactions. We iteratively remove sessions shorter than 5 and items with fewer than 3 interactions until convergence, then split sessions chronologically into training, validation, and test partitions. The held-out tails are re-formed into contiguous validation and test segments, and non-target unseen rows are removed while unseen targets are preserved. We report HR, NDCG, and MRR at $k \in \{5, 10, 20\}$ for all methods, and use validation MRR@20 for model selection when a single scalar criterion is required.

A.2 Full Ranking Results

Table 3 restores the full cutoff grid that is summarized more compactly in Table 1. This presentation is useful for checking whether the RouteRec advantage is confined to one evaluation cutoff or remains visible across shallower and deeper retrieval depth.

The appendix table makes it easier to confirm whether a gain survives beyond the single-cutoff summary used in the main paper. In the final manuscript, this table can also absorb any dataset-specific notes about variance or significance without interrupting the narrative of the main experiments section.



(a) Macro/mid/micro removal (b) Dense FFN vs. 1/2/3-stage layouts

(c) Wrapper-map and stage-order variants

Figure 6: Extended stage and layout ablations. (a) Stage removal. (b) Dense replacement versus staged routing. (c) Wrapper-map and stage-order variants.

B Extended Stage and Layout Ablations

Figure 6 expands the structural study in Figure 4 with the broader sweep of stage-removal and wrapper variants.

The figure should be read from left to right. Panel (a) verifies that macro, mid, and micro routing each contribute a distinct part of the gain. Panel (b) separates staged routing from generic expert insertion, and panel (c) checks that the final wrapper map is not an arbitrary configuration choice.

C Objective and Regularization Variants

Table 4 isolates the contribution of consistency, z-loss, and balance regularization using ranking quality, route consistency, and stability indicators.

This compact table shows whether the full objective improves MRR@20 while preserving routing consistency and stability.

D Extended Diagnostics

Figure 7 groups the main routing diagnostics into one coordinated layout so that stage usage, sharpness, and semantic consistency can be read together.

These diagnostics are organized from coarse structure to semantic interpretation. The first two panels show how broad or sharp routing is at each stage, while the last two panels connect those patterns back to feature similarity and interpretable behavioral regimes.

Dataset	Metric	GRU4Rec	SASRec	TiSASRec	DuoRec	FEARec	BSARec	DIF-SR	SIGMA	RouteRec	Improv.
Beauty	HR@5	0.0681	0.0834	0.0786	0.0915	0.0952	0.0897	0.0813	0.0869	**0.0988**	+3.8%
	HR@10	0.0925	0.1082	0.1016	0.1187	0.1219	0.1168	0.0994	0.1048	**0.1256**	+3.0%
	HR@20	0.1220	0.1395	0.1324	0.1510	0.1548	0.1491	0.1282	0.1344	**0.1589**	+2.6%
	NDCG@5	0.0517	0.0635	0.0592	0.0694	0.0717	0.0672	0.0598	0.0621	**0.0745**	+3.9%
	NDCG@10	0.0592	0.0714	0.0667	0.0778	0.0798	0.0750	0.0661	0.0693	**0.0826**	+3.5%
	NDCG@20	0.0665	0.0791	0.0742	0.0856	0.0871	0.0824	0.0734	0.0762	**0.0896**	+2.9%
	MRR@5	0.0480	0.0587	0.0544	0.0648	0.0661	0.0625	0.0553	0.0580	**0.0687**	+3.9%
	MRR@10	0.0524	0.0639	0.0592	0.0698	0.0713	0.0670	0.0598	0.0627	**0.0739**	+3.6%
	MRR@20	0.0548	0.0667	0.0619	0.0727	0.0741	0.0698	0.0623	0.0651	**0.0763**	+3.0%
Foursquare	HR@5	0.1432	0.1964	0.1861	0.1937	0.1820	0.1681	0.1734	0.1885	**0.2049**	+4.3%
	HR@10	0.1827	0.2361	0.2210	0.2335	0.2178	0.2050	0.2106	0.2274	**0.2447**	+3.6%
	HR@20	0.2294	0.2850	0.2687	0.2829	0.2661	0.2525	0.2578	0.2744	**0.2939**	+3.1%
	NDCG@5	0.0983	0.1396	0.1314	0.1365	0.1277	0.1188	0.1220	0.1342	**0.1457**	+4.4%
	NDCG@10	0.1111	0.1527	0.1439	0.1494	0.1396	0.1301	0.1330	0.1464	**0.1589**	+4.1%
	NDCG@20	0.1230	0.1655	0.1564	0.1620	0.1517	0.1419	0.1443	0.1583	**0.1718**	+3.8%
	MRR@5	0.0887	0.1282	0.1201	0.1244	0.1157	0.1085	0.1102	0.1228	**0.1339**	+4.4%
	MRR@10	0.0945	0.1344	0.1261	0.1305	0.1219	0.1140	0.1162	0.1287	**0.1400**	+4.2%
	MRR@20	0.0974	0.1378	0.1294	0.1336	0.1248	0.1168	0.1191	0.1315	**0.1430**	+3.8%
KuaiRec	HR@5	0.1346	0.1412	0.1279	0.1434	0.1422	0.1397	0.1331	0.1406	**0.1451**	+1.2%
	HR@10	0.1732	0.1798	0.1647	0.1809	0.1796	0.1764	0.1691	0.1778	**0.1832**	+1.3%
	HR@20	0.2195	0.2258	0.2094	0.2261	0.2251	0.2213	0.2146	0.2234	**0.2287**	+1.2%
	NDCG@5	0.1043	0.1109	0.0988	0.1116	0.1107	0.1084	0.1023	0.1095	**0.1138**	+2.0%
	NDCG@10	0.1169	0.1236	0.1108	0.1241	0.1230	0.1208	0.1137	0.1216	**0.1264**	+1.9%
	NDCG@20	0.1286	0.1351	0.1222	0.1352	0.1343	0.1320	0.1249	0.1332	**0.1376**	+1.8%
	MRR@5	0.0987	0.1049	0.0925	0.1056	0.1044	0.1027	0.0969	0.1035	**0.1079**	+2.2%
	MRR@10	0.1056	0.1118	0.0994	0.1123	0.1110	0.1094	0.1031	0.1100	**0.1147**	+2.1%
	MRR@20	0.1098	0.1160	0.1036	0.1164	0.1152	0.1133	0.1070	0.1141	**0.1185**	+1.8%
LastFM	HR@5	0.3221	**0.3604**	0.2415	0.2549	0.2627	0.3492	0.2443	0.3528	0.3561	-1.2%
	HR@10	0.4016	**0.4456**	0.2564	0.2694	0.2774	0.4280	0.2585	0.4259	0.4305	-3.4%
	HR@20	0.4928	**0.5369**	0.2849	0.2975	0.3041	0.5246	0.2868	0.5208	0.5282	-1.6%
	NDCG@5	0.2714	**0.3048**	0.2017	0.2079	0.2124	0.2958	0.2033	0.3002	0.3021	-0.9%
	NDCG@10	0.3104	**0.3428**	0.2178	0.2236	0.2282	0.3345	0.2197	0.3383	0.3391	-1.1%
	NDCG@20	0.3445	**0.3771**	0.2320	0.2370	0.2411	0.3702	0.2337	0.3730	0.3748	-0.6%
	MRR@5	0.2582	**0.2901**	0.1904	0.1952	0.1996	0.2827	0.1919	0.2869	0.2881	-0.7%
	MRR@10	0.2726	**0.3040**	0.2019	0.2064	0.2107	0.2971	0.2032	0.3004	0.3015	-0.8%
	MRR@20	0.2801	**0.3114**	0.2082	0.2126	0.2168	0.3045	0.2094	0.3071	0.3083	-1.0%
ML-1M	HR@5	0.1112	**0.1265**	0.1258	0.1087	0.1183	0.1227	0.1235	0.1190	0.1218	-3.7%
	HR@10	0.1694	**0.1887**	0.1878	0.1652	0.1790	0.1841	0.1850	0.1790	0.1826	-3.2%
	HR@20	0.2426	**0.2683**	0.2674	0.2360	0.2509	0.2617	0.2631	0.2536	0.2585	-3.7%
	NDCG@5	0.0716	**0.0804**	0.0800	0.0678	0.0734	0.0772	0.0781	0.0745	0.0760	-5.5%
	NDCG@10	0.0908	0.0952	0.0953	0.0838	0.0869	0.0983	**0.0990**	0.0936	0.0978	-1.2%
	NDCG@20	0.1094	0.1137	0.1139	0.1011	0.1046	0.1186	**0.1194**	0.1125	0.1170	-2.0%
	MRR@5	0.0648	**0.0720**	0.0718	0.0615	0.0641	0.0684	0.0695	0.0657	0.0678	-5.8%
	MRR@10	0.0689	**0.0761**	0.0760	0.0643	0.0664	0.0719	0.0728	0.0687	0.0712	-6.4%
	MRR@20	0.0711	**0.0783**	0.0781	0.0659	0.0679	0.0747	0.0756	0.0719	0.0741	-5.4%
Retail Rocket	HR@5	0.2894	0.3368	0.3207	0.3389	0.3371	0.3314	0.3345	0.3168	**0.3569**	+5.3%
	HR@10	0.3779	0.4391	0.4205	0.4427	0.4418	0.4327	0.4400	0.4155	**0.4723**	+6.7%
	HR@20	0.4682	0.5364	0.5188	0.5411	0.5390	0.5284	0.5379	0.5072	**0.5756**	+6.4%
	NDCG@5	0.1981	0.2327	0.2214	0.2348	0.2339	0.2262	0.2311	0.2147	**0.2484**	+5.8%
	NDCG@10	0.2504	0.2907	0.2798	0.2921	0.2915	0.2870	0.2896	0.2748	**0.3094**	+5.9%
	NDCG@20	0.2760	0.3165	0.3053	0.3182	0.3171	0.3110	0.3137	0.2984	**0.3361**	+5.6%
	MRR@5	0.1765	0.2088	0.1995	0.2104	0.2098	0.2031	0.2075	0.1922	**0.2230**	+6.0%
	MRR@10	0.1918	0.2260	0.2156	0.2274	0.2267	0.2194	0.2241	0.2083	**0.2410**	+6.0%
	MRR@20	0.2001	0.2348	0.2241	0.2362	0.2354	0.2278	0.2327	0.2163	**0.2495**	+5.6%

Table 3: Full ranking results across all cutoffs.

E Behavior-Slice Analysis

Because RouteRec is motivated by heterogeneous behavior, a slice-wise view is a natural complement to the aggregate metrics. This appendix section is intended to localize gains to behavior regimes such as repeat-heavy prefixes, high-tempo sessions, narrow-focus sessions, and exploration-heavy prefixes.

The useful reading order is to first check which slices benefit most in panel (a), then verify in panel (b) whether those gains coincide with more decisive and semantically aligned routing. This experiment fits the paper's main argument especially well because it answers not only whether RouteRec helps, but also where the help appears.

Table 4: Objective and regularization variants.

Variant	MRR@20	Consistency	Stability
No auxiliary loss	–	–	–
KNN + consistency	–	–	–
z-loss only	–	–	–
Balance loss only	–	–	–
Consistency + z-loss	–	–	–
Full objective	–	–	–

(a) Expert-usage heatmaps by stage

(b) Entropy and effective-expert-count summary

(c) Stage-wise route consistency

(d) Feature-bucket route patterns

Figure 7: Routing diagnostics. (a) Expert usage by stage. (b) Entropy and effective expert count. (c) Stage-wise route consistency. (d) Feature-bucket routing patterns.

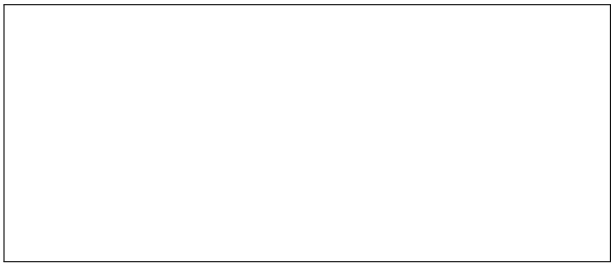
Table 5: Source-target transfer matrix.

Src. → Tgt.	Bty	Fsq	KRec	LFM	ML1M	RR
Bty	–	–	–	–	–	–
Fsq	–	–	–	–	–	–
KRec	–	–	–	–	–	–
LFM	–	–	–	–	–	–
ML1M	–	–	–	–	–	–
RR	–	–	–	–	–	–

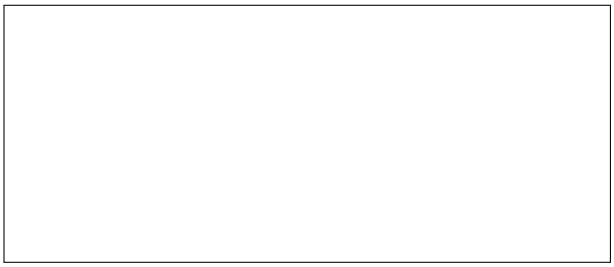
F Extended Transfer Results

This section reports additional transfer experiments for low-resource and cross-domain adaptation.

The matrix provides the exact source-target view that is too detailed for the main paper. It is useful for identifying asymmetric transfer pairs and for checking whether router reuse is most effective between domains with similar temporal and repetition structure. For compactness, the table uses the abbreviations Bty, Fsq, KRec, LFM, ML1M, and RR.

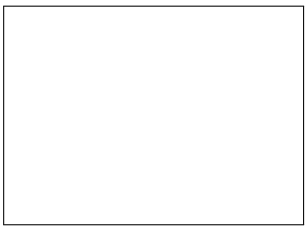


(a) Slice-wise MRR@20 across behavioral regimes

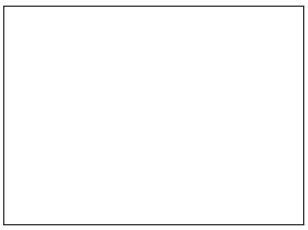


(b) Relative RouteRec gain and route concentration by slice

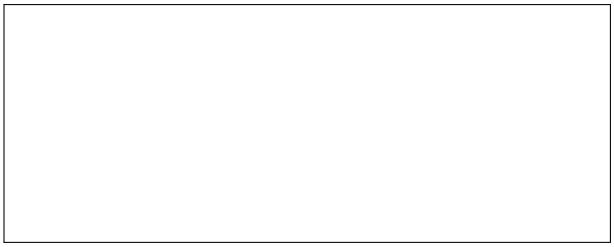
Figure 8: Behavior-slice analysis. (a) Slice-wise ranking quality. (b) Relative gain and routing concentration by slice.



(a) Low-resource transfer curves



(b) Frozen/finetuned/anchor and group/full router transfer



(c) Multi-source sequential order effect

Figure 9: Transfer variants. (a) Low-resource transfer curves. (b) Frozen, finetuned, anchor, and router-granularity comparisons. (c) Multi-source order effects.

Together, the table and figure let the reader move from exact pairwise results to higher-level patterns. The intended appendix message is that transfer is asymmetric, most helpful in related domains and low-resource regimes, and partially preserved even under coarser router reuse.